Dummy Variable Creation & Encoding – Explanation

This code converts categorical variables (Gender and Education Level) into numerical form so they can be used in regression analysis.

Regression models cannot understand text values like “male” or “bachelors”, so we convert them into numbers.

STEP 1: Load Cleaned Dataset

df = pd.read_csv("cleaned_salary_dataset_v1.csv")

- Loads the already cleaned dataset.
- We use cleaned data instead of raw data to avoid errors.
- Displays columns to confirm correct structure.

Purpose:
To begin encoding using a properly prepared dataset.

STEP 2: Encode Gender (Binary Encoding)

df['gender'] = df['gender'].map({'male': 1, 'female': 0})

What This Does:

It converts:

Gender	Encoded Value
male	1
female	0

This is called **binary dummy encoding**.

Why This Is Important:

- Regression requires numeric values.
- Now the model can measure salary difference between male and female.

How It Works in Regression:

If gender coefficient = 5000
→ Males earn ₹5000 more than females (since male = 1).

If gender coefficient = -3000
→ Females earn ₹3000 more than males.

So this variable directly helps measure the gender pay gap.

STEP 3: One-Hot Encode Education Level

df = pd.get_dummies(df, columns=['education_level'], drop_first=True)

Education level has multiple categories like:

- high_school
- bachelors
- masters
- phd

Since it has more than two categories, we use **one-hot encoding**.

What One-Hot Encoding Does

Suppose original categories are:

- high_school
- bachelors
- masters
- phd

After encoding (with drop_first=True), we may get:

education_level_bachelors	education_level_masters	education_level_phd
0	0	0
1	0	0
0	1	0
0	0	1

Here:

- high_school becomes the reference category.
- Other levels are compared against it.

Why drop_first=True?

If we keep all categories, it causes **Dummy Variable Trap** (perfect multicollinearity).

By dropping one category:

- We avoid redundancy.
- We keep one category as reference group.
- Regression becomes mathematically valid.

Interpretation in Regression

If:

education_level_masters coefficient = 8000
→ Masters graduates earn ₹8000 more than high school graduates.

If:

education_level_phd coefficient = 15000
→ PhD holders earn ₹15000 more than high school graduates.

STEP 4: Save Final Encoded Dataset

df.to_csv("final_encoded_dataset.csv", index=False)

This saves the fully encoded dataset.

Now the dataset is ready for:

- Building regression model
- Statistical analysis
- Gender pay gap testing

Why This Step Is Very Important in Your Project

This encoding allows you to:

✓ Include gender in regression
✓ Compare education levels
✓ Measure salary difference numerically
✓ Avoid multicollinearity
✓ Detect unexplained gender pay gap

Without dummy variables, regression analysis is impossible for categorical data.

"We converted categorical variables into dummy variables because regression models require numeric inputs. Gender was binary encoded as male = 1 and female = 0. Education level was one-hot encoded with drop_first=True to avoid dummy variable trap. This allowed us to measure salary differences across gender and education levels accurately."